E-Cigarette Supply — Solution Appendix

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Draft

Full equation systems and solution algorithm, conference version. Companion to E-Cigarette Supply: A Conference-Version Model*, which has the motivation and the model stated at paper-length detail — this note exists to show exactly what gets solved and how.*

0. Scope

Notation matches the companion model note: period t, active incumbents $i = 1, \ldots, n_t \leq \bar{N} = 5$, a fringe F, at most one potential entrant, state $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$, both exogenous in this version. This note works through three solved objects in order: (1) the static pricing subgame, as an explicit simultaneous system; (2) the state space and its transition; (3) the dynamic entry/exit program, as an explicit nested fixed point — then closes with the outer algorithm that ties them together for a given parameter vector.

1 Demand system

Firm i's mean utility: $\delta_{i,t} = \beta_a a_{i,t} + \xi_{i,t} - \alpha p_{i,t}$, where β_a is a free coefficient on the quality state (not normalized to 1 — see the companion model note §3 for the Ericson–Pakes 1994 precedent) and $\xi_{i,t}$ is the exogenous residual carried in $\omega_{i,t} = (a_{i,t}, \xi_{i,t})$, included per standard mean-utility convention (Berry 1994/BLP: mean utility is the part of utility common across consumers, not just what's observed by the econometrician). All active incumbents and the fringe sit in one nest g; the outside option is its own singleton nest, mean utility normalized to 0. With nest parameter $\sigma \in [0, 1)$:

$$s_{i|g,t} = \frac{\exp\big(\delta_{i,t}/(1-\sigma)\big)}{D_{g,t}}, \qquad D_{g,t} = \sum_{j \in g,t} \exp\big(\delta_{j,t}/(1-\sigma)\big) \tag{1}$$

$$IV_{g,t} = (1-\sigma)\,\ln D_{g,t} \tag{2}$$

$$s_{g,t} = \frac{\exp(IV_{g,t})}{1 + \exp(IV_{g,t})} \qquad \text{[prob. of choosing the nest at all]} \tag{3}$$

$$s_{i,t} = s_{i|g,t} \cdot s_{g,t} \qquad \text{[unconditional share]} \tag{4}$$

Price and data construction (7 October 2026). Each UPC-month price is real revenue divided by corrected mL sold; $p_{i,t}$ is the unweighted mean over eligible UPCs (positive sales, at least three stores, UNKNOWN excluded). Reviewed pack sizes/counts feed retail volume, nicotine and shares. NCP purchasing-intensity inputs remain unchanged. $p_{i,t}$ times firm mL need not equal observed revenue.

Diagnostic controls: unweighted cartridge/disposable capacity and log UPC count; report missing-size exclusions. Compare annual, monthly and constant NCP intensity times annual adult population. Current estimates and instrument specifications are in the comparison report; they do not establish a valid supply-side calibration.

Price derivatives (standard nested-logit result — e.g. Train, *Discrete Choice Methods with Simulation*, ch. 4; worth checking against the primary source before this goes in a draft). For

i, j both in nest g:

$$\frac{\partial s_i}{\partial p_i} = -\alpha\, s_i \left[\frac{1}{1-\sigma} - \frac{\sigma}{1-\sigma}\, s_{i|g} - s_i \right] \qquad \text{[own price]} \tag{5}$$

$$\frac{\partial s_i}{\partial p_j} = \alpha\, s_i \left[\frac{\sigma}{1-\sigma}\, s_{j|g} + s_j \right] \qquad j \neq i,\ j \in g \qquad \text{[rival's price]} \tag{6}$$

These reduce to the standard multinomial-logit derivatives at $\sigma = 0$ ($\partial s_i/\partial p_i \to -\alpha\, s_i(1-s_i)$, $\partial s_i/\partial p_j \to \alpha\, s_i s_j$), which is a useful check when this gets coded up.

Demand estimation, step by step. Eqs. (1)–(4) are a model of shares, not yet an estimation procedure — this walks through how $(\beta_a, \alpha, \sigma, \xi_{i,t})$ are actually recovered from data. Two things drive the procedure: the data (Nielsen RMS) gives aggregate market shares, not individual consumer choices, so a choice-level MLE (e.g. Stata's `nlogit`) doesn't apply here; and the nested logit's share formula happens to have a special property (Berry 1994) — it can be inverted and linearized — that turns estimation into an IV/2SLS regression instead.

1. **Construct shares from data.** For each incumbent i and period t, $s_{i,t}$ = (firm i's volume) / (total market size, where market size includes every incumbent, the fringe, and the outside option). Construct $s_{F,t}$ the same way for the fringe (one aggregate row per period, not broken out by sub-cap brand), and $s_{0,t} = 1 - \sum_i s_{i,t} - s_{F,t}$ for the outside option. $s_{g,t} = \sum_i s_{i,t} + s_{F,t}$ is the total nest share — the fringe's volume belongs in this sum because it competes inside nest g (§6 of the model note).

2. **Build the estimating equation's two sides — no parameters needed yet.** For each incumbent (i,t), compute the dependent variable $\ln(s_{i,t}) - \ln(s_{0,t})$ and the within-nest-share regressor $\ln(s_{i,t}/s_{g,t})$. Both are pure functions of observed shares — σ doesn't need to be known to compute either one.

$$\ln(s_{i,t}) - \ln(s_{0,t}) \;=\; \beta_a a_{i,t} \;-\; \alpha p_{i,t} \;+\; \sigma \ln(s_{i,t}/s_{g,t}) \;+\; \xi_{i,t} \tag{7}$$

3. **Run this as one IV/2SLS regression, incumbents only.** $a_{i,t}$ is data (the UPC-unweighted nicotine construction, §6 of the model note); $p_{i,t}$ and $\ln(s_{i,t}/s_{g,t})$ are both instrumented, since both are correlated with $\xi_{i,t}$ (firms price higher, and capture more within-nest share, exactly when their unobserved quality $\xi_{i,t}$ is high — the standard simultaneity problem). $\hat{\beta}_a$, $\hat{\alpha}$, and $\hat{\sigma}$ all come out of this single regression as coefficients — σ is never chosen or plugged in beforehand; it's estimated jointly with the other two. $\hat{\xi}_{i,t}$ is the regression's residual, not an input.

4. **The fringe enters step 1, not step 3.** Its volume is already inside $s_{g,t}$ and $s_{0,t}$ for every incumbent observation, so it's already shaping every coefficient — but it is never its own row in the regression, because this specification gives it no price and no $\xi_{F,t}$ term to identify. $a_{F,t}$ is constructed directly from data, the same way as $a_{i,t}$ (an aggregate/composite nicotine measure across the sub-cap brands, per the model note §4.5) — it is not backed out from the demand estimation. One consequence worth flagging: because the fringe has no $\xi_{F,t}$ to absorb error, $\hat{\beta}_a a_{F,t}$ generally will not exactly reproduce the fringe's observed $s_{F,t}$ — unlike every incumbent, which fits exactly by construction (that's what $\xi_{i,t}$ is for).

A notational note, since it's easy to conflate. $\xi_{i,t}$ here is not the same object as the ε_{ijt} you may be used to from individual-level discrete-choice models ($u_{ijt} = \delta_{j,t} + \varepsilon_{ijt}$, ε_{ijt} iid Type I Extreme Value across person i, alternative j, and time t). ε_{ijt} is a modeling assumption about person-level taste heterogeneity — it's integrated out analytically to derive the share formulas (1)–(4) in the first place, and it's never estimated. $\xi_{i,t}$ has no person subscript at all: it's the aggregate, firm-and-time-level residual left over after fitting the mean-utility equation to market-level share data, recovered because the econometrician is missing information about the *product*, not about any individual consumer.

2 Static pricing subgame

Firm i's problem, given the period's state ω_t and taking rivals' prices as given:

$$\max_{p_i} \;\; \pi_i(p) = (p_i - c_i)\, q_i(p) - FC\,, \qquad c_i \equiv c(\omega_{i,t}), \qquad q_i(p) \equiv M_t \cdot s_i(p). \tag{8}$$

$q_i(p)$ is quantity; $s_i(p)$ is the market share from (4), and M_t is market size.

First-order condition ($q_i + (p_i - c_i)\,\partial q_i/\partial p_i = 0$; since $q_i(p) = M_t\, s_i(p)$, dividing through by $M_t > 0$ leaves the FOC purely in shares — market size drops out, substituting (5)):

$$p_i = c_i \;+\; \frac{1}{\alpha\Big[\dfrac{1}{1-\sigma} - \dfrac{\sigma}{1-\sigma}\, s_{i|g}(p) - s_i(p)\Big]} \qquad \text{for every active } i = 1, \ldots, n_t. \tag{9}$$

This is the object to solve: n_t **simultaneous nonlinear equations in n_t unknowns** $(p_1, \ldots, p_{n_t})$, since the right-hand side of each depends on the full price vector through $s_i(p)$, $s_{i|g}(p)$. The fringe is *not* an extra unknown in this version — its mean utility $\delta_{F,t} = \beta_a a_{F,t}$ is exogenous, so its share $s_F(p)$ enters the $D_{g,t}$ denominator (and hence every firm's shares) as a known quantity, not something solved for. (§4 of the model note; a strategic fringe would add one more equation of the form (9) to this system.)

Solving it — fixed-point iteration. Guess $p^{(0)}$ (e.g. c_i plus a small markup for each i); iterate

$$p_i^{(k+1)} \;=\; c_i \;+\; \frac{1}{\alpha\Big[\dfrac{1}{1-\sigma} - \dfrac{\sigma}{1-\sigma}\, s_{i|g}\big(p^{(k)}\big) - s_i\big(p^{(k)}\big)\Big]} \qquad \text{for all } i, \tag{10}$$

recomputing shares at each step, until $\max_i \big|p_i^{(k+1)} - p_i^{(k)}\big| < \text{tol}$. This "best-response" / successive-substitution iteration is the standard first pass; it is not guaranteed to contract in every logit specification, so a production solve should fall back to Newton's method on the stacked FOC system (9), or the ζ-markup reformulation of Morrow & Skerlos (2011), if (10) fails to converge or converges slowly.

Worked example — two incumbents plus the (passive) fringe. With $n_t = 2$ (firms 1, 2) and fringe F contributing a fixed $\delta_{F,t}$:

$$D_g = \exp\big(\delta_1/(1-\sigma)\big) + \exp\big(\delta_2/(1-\sigma)\big) + \exp\big(\delta_F/(1-\sigma)\big)$$

$$s_{1|g} = \frac{\exp(\delta_1/(1-\sigma))}{D_g}\,, \qquad s_{2|g} = \frac{\exp(\delta_2/(1-\sigma))}{D_g}$$

$$IV_g = (1-\sigma)\ln D_g\,, \qquad s_g = \frac{\exp(IV_g)}{1+\exp(IV_g)}$$

$$s_1 = s_{1|g} \cdot s_g\,, \qquad s_2 = s_{2|g} \cdot s_g$$

$$\text{FOC}_1: \quad p_1 = c_1 + \frac{1}{\alpha\big[\,1/(1-\sigma) - (\sigma/(1-\sigma))\,s_{1|g} - s_1\,\big]}$$
$$\text{FOC}_2: \quad p_2 = c_2 + \frac{1}{\alpha\big[\,1/(1-\sigma) - (\sigma/(1-\sigma))\,s_{2|g} - s_2\,\big]}$$

Two equations, two unknowns (p_1, p_2) — $\delta_1, \delta_2, \delta_F, \alpha, \sigma, c_1, c_2$ all fixed inputs for this state. This is exactly what gets solved, at every state ω_t the dynamic program visits, for every candidate parameter vector.

3 State space and transition

The industry state is the set of active firms and their individual states, $\Omega_t = \{\,\omega_{i,t} : i \text{ active at } t\,\}$, with $n_t \le \bar{N}$. Practically, $a_{i,t}$ and $\xi_{i,t}$ are each discretized onto a finite grid (integers, per the demand-side mean-utility construction), so Ω lives on a finite (if large) grid.

Transition, conditional on actions. For a continuing incumbent i (did not exit):

$$a_{i,t+1} \sim F_a(\,\cdot \mid a_{i,t}\,), \qquad \xi_{i,t+1} \sim F_\xi(\,\cdot \mid \xi_{i,t}\,) \qquad \text{[independent, exogenous]} \tag{11}$$

For an exiting incumbent: i leaves the active set permanently (absorbing). For an entering firm: it joins the active set at $t+1$ with an initial state $\omega_{e,0}$ drawn from the estimated entrant-state distribution $G_e(\cdot)$ — note the "0" subscript denotes age-since-entry (the entrant's own initial draw), not calendar time; $\omega_{e,0}$ is distinct in kind from $\omega_{i,t}/\omega_{i,t+1}$, which track an incumbent's realized history. The full transition kernel $\Pr(\Omega_{t+1} \mid \Omega_t, \text{exit/entry decisions})$ is the product of (11) across continuing firms, combined with the deterministic removal of exiters and the (possibly probabilistic, if entry itself is uncertain conditional on the decision to try) addition of an entrant drawn from $G_e(\cdot)$.

Constructing F_a and F_ξ, step by step. (11) states that these kernels exist; this is how they're actually built — an empirical first-order Markov transition matrix, the same technique for both, applied to two different series.

1. **Put each variable on a finite grid.** $a_{i,t}$ is already close to this by construction (the UPC-unweighted nicotine measure, §6 of the model note). $\hat{\xi}_{i,t}$ is not — it comes out of §1's demand regression as a continuous residual, so it needs to be binned onto a finite grid (evenly-spaced or quantile bins) before this procedure applies to it. This also means F_ξ can only be built *after* the demand estimation step, since $\hat{\xi}_{i,t}$ doesn't exist until that regression has run.

2. **Keep only continuing-incumbent transitions.** Restrict to $(i,t) \to (i,t+1)$ pairs where firm i is active in both periods — matching (11)'s own scope. An exiting firm contributes no transition; an entering firm's initial state is handled separately by $G_e(\cdot)$, not by F_a/F_ξ.

3. **Pool across firms and count.** For each grid value a, count how many pooled $(i,t) \to (i,t+1)$ pairs have $a_{i,t} = a$ and $a_{i,t+1} = a'$. This assumes one common transition process across firms, not a firm-specific one — the standard simplification here.

4. **Normalize each row.**

$$F_a(a' \mid a) \;=\; \frac{\#\{\text{pairs with } a_{i,t} = a,\ a_{i,t+1} = a'\}}{\#\{\text{pairs with } a_{i,t} = a\}} \tag{12}$$

Repeat the identical procedure on the binned $\hat{\xi}_{i,t}$ series to get F_ξ.

5. **Watch for sparse cells.** With an 11-year panel and 5–7 firms, a fine grid can leave some (a, a') cells with very few or zero observations, making those transition probabilities noisy or undefined. If that happens: coarsen the grid, or fall back to a semi-parametric approach — fit a continuous AR(1)-type law of motion and discretize it onto a grid via Tauchen's method, trading some flexibility for a fully-populated matrix. F_F (the fringe's own transition, §4.5 of the model note) is built the same way, once $a_{F,t}$ itself is constructed.

4 Dynamic entry/exit program

Incumbent's Bellman equation. With $\pi_i^*(\omega_t)$ the equilibrium static profit from §2 at state ω_t:

$$V_i(\omega_{i,t}, \omega_{-i,t}) \;=\; \pi_i^*(\omega_t) \;+\; \max\Big\{ \underbrace{\phi_i}_{\text{exit: scrap value}} \;,\; \underbrace{\beta \cdot \mathbb{E}\big[\, V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue} \big]}_{\text{continue}} \Big\} \tag{13}$$

Optimal exit rule: firm i exits iff $\phi_i > \beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}]$ — a cutoff in $\omega_{i,t}$ given rivals' states, since the continuation value is (weakly) increasing in the firm's own state. Define the exit indicator implied by this cutoff:

$$\lambda^{\text{exit}}(\omega_{i,t}; \omega_{-i,t}) \;=\; \mathbf{1}\big\{\, \phi_i > \beta \cdot \mathbb{E}[V_i(\omega_{i,t+1}, \omega_{-i,t+1}) \mid \omega_t, \text{continue}] \,\big\} \tag{14}$$

(λ, not η — in Ericson–Pakes' own model η' already denotes something unrelated, an aggregate industry-wide shock inside the transition kernel, and χ is claimed too, as the exit-decision indicator itself in the closely-related Doraszelski–Pakes–Satterthwaite restatement of this model. λ sidesteps both of those as well as the nested-logit correlation parameter σ used in §1–§2.)

Entrant's problem. The potential entrant observes the current state and enters iff its expected discounted value of entry, net of the sunk entry cost, is non-negative:

$$\text{enter} \iff \beta \cdot \mathbb{E}\big[\, V_e(\omega_{e,0}, \omega_{-e,t+1}) \mid \omega_t \,\big] \;-\; x_e \;\geq\; 0 \tag{15}$$

where $\omega_{e,0}$ is the entrant's initial-state draw from the estimated entrant-state distribution $G_e(\cdot)$ (§3), and V_e is evaluated using the same value function as incumbents (an entering firm becomes an ordinary incumbent from $t+1$ on). Define the entry indicator implied by this cutoff:

$$\lambda^{\text{entry}}(\omega_t) \;=\; \mathbf{1}\big\{\, \beta \cdot \mathbb{E}[V_e(\omega_{e,0}, \omega_{-e,t+1}) \mid \omega_t] - x_e \geq 0 \,\big\} \tag{16}$$

Solving it — nested fixed point (Pakes–McGuire 1994). For a fixed set of primitives ($\pi_i^*(\cdot)$ from §2 at every state, F_a, F_ξ, G_e, ϕ_i, x_e, β), solve for $(V_i, \text{exit rule}, \text{entry rule})$ jointly:

Algorithm 1 Nested fixed point for the entry/exit equilibrium

Given: static profits $\{\pi_i^*(\omega)\}$ over the state grid; primitives $F_a, F_\xi, G_e, \phi_i, x_e, \beta$

guess exit rule λ^{exit} and entry rule λ^{entry}
repeat ▷ **Outer loop** — fixed point in policies
  guess $V_i^{(0)}(\omega)$ over the state grid
  repeat ▷ **Inner loop** — value function iteration, given $(\lambda^{\text{exit}}, \lambda^{\text{entry}})$
    for every state ω **do**
      $V_i^{(k+1)}(\omega) \leftarrow \pi_i^*(\omega) + \max\left\{ \phi_i\,,\ \beta \sum_{\omega'} \Pr(\omega' \mid \omega, \lambda^{\text{exit}}, \lambda^{\text{entry}})\, V_i^{(k)}(\omega') \right\}$
    end for
  until $\max_\omega \left| V_i^{(k+1)}(\omega) - V_i^{(k)}(\omega) \right| < \text{tol}_V$ ▷ contraction in $\beta < 1$: guaranteed
  update λ^{exit} from the converged V_i via the cutoff rule under (13)
  update λ^{entry} from the converged V_i via (15)
until both policies are unchanged from the previous outer iteration ▷ no general guarantee

This is the literal structure behind "conditional on the Bellman, iterate on best response; conditional on best response, update the value function" — the inner loop is a standard contraction-mapping value-function iteration (guaranteed to converge because $\beta < 1$); the outer loop is the fixed point in policies that makes this a genuine equilibrium rather than just a single agent's optimization, and is the piece with no general convergence guarantee — Pakes & McGuire (2001)'s contribution is restricting the state grid the inner loop iterates over to the recurrent set actually reached under simulated play, which is what keeps this tractable as the state space grows (see the companion model note, §3, on why that alone stops being enough once investment is added, motivating EBE for the full model).

What $\Pr(\omega' \mid \omega, \lambda^{\text{exit}}, \lambda^{\text{entry}})$ actually is — a worked example. Take a 2-incumbent industry (firms 1 and 2) plus one potential entrant, state $\omega = (\omega_1, \omega_2)$. λ^{exit} and λ^{entry} are deterministic given ω — they say *who's on the roster* next period, not where on the grid each surviving/entering firm lands:

- If $\lambda^{\text{exit}}(\omega_1; \omega_2) = 0$ and $\lambda^{\text{exit}}(\omega_2; \omega_1) = 0$ (both continue) and $\lambda^{\text{entry}}(\omega) = 0$ (no entry), next period's roster is $\{1, 2\}$ for sure — any ω' with a different roster (e.g. one with only firm 1, or with a third firm) gets probability exactly 0.
- Given that roster, F_a and F_ξ (independently for each continuing firm, per (11)) supply the actual probability weight over *where* each firm's state lands: $\Pr(\omega_1', \omega_2' \mid \omega) = F_a(a_1' \mid a_1)\, F_\xi(\xi_1' \mid \xi_1) \cdot F_a(a_2' \mid a_2)\, F_\xi(\xi_2' \mid \xi_2)$.
- If instead $\lambda^{\text{exit}}(\omega_1; \omega_2) = 1$ (firm 1 exits) and $\lambda^{\text{entry}}(\omega) = 1$ (entry happens), next period's roster is $\{2, \text{entrant}\}$ for sure, and the probability weight over ω' comes from $F_a(\cdot \mid a_2)\, F_\xi(\cdot \mid \xi_2)$ for firm 2 combined with $G_e(\cdot)$ for the entrant's fresh draw $\omega_{e,0}$ — firm 1 contributes nothing to next period's state at all.

So $\Pr(\omega' \mid \omega, \lambda^{\text{exit}}, \lambda^{\text{entry}})$ is the product of two pieces: the (deterministic, 0/1) roster implied by $\lambda^{\text{exit}}/\lambda^{\text{entry}}$, and the (random) $F_a/F_\xi/G_e$ draws that place the surviving/entering firms on the grid conditional on that roster.

5 Putting it together — the outer algorithm

For a candidate parameter vector $\theta = (\alpha, \sigma, \text{cost-function parameters}, \phi, x_e, F_a, F_\xi, \dots)$:

Algorithm 2 Full solve for a given parameter vector θ

1: **for** every state ω on the grid **do**
2: solve the static pricing system (9)–(10) $\rightarrow$ get $p^*(\omega)$ and $\pi_i^*(\omega)$ for every active i
3: **end for**
4: using $\{\pi_i^*(\omega)\}$: solve the dynamic program of §4 (Algorithm 1) to convergence $\rightarrow$ get $V_i(\cdot)$, the exit rule, the entry rule
5: simulate the model forward under the converged policies (or compute the model's implied moments directly) for whatever estimation routine θ is being chosen to match

Step 1 is re-solved for every θ only insofar as marginal costs or δ depend on θ; if the pricing-relevant parameters $(\alpha, \sigma, c(\cdot))$ are held fixed while only the dynamic parameters (ϕ, x_e, transition parameters) are being searched over, step 1's results can be cached and reused across outer iterations — worth exploiting computationally once this is actually being estimated, since step 1 (a fixed point in prices, at every grid state) is the part most likely to dominate runtime otherwise.

Appendix — notation

Symbol	Meaning
t	period (month)
i, j	active firms, $i, j = 1, \ldots, n_t \leq \bar{N} = 5$
F	the fringe (single aggregated player)
$\omega_{i,t} = (a_{i,t}, \xi_{i,t})$	firm i's state: accumulated-quality index, exogenous residual
$a_{F,t}$	fringe's aggregate state
$\delta_{i,t}$	mean utility, $= \beta_a a_{i,t} + \xi_{i,t} - \alpha p_{i,t}$
β_a	free coefficient on the quality state $a_{i,t}$ in mean utility (not normalized to 1)
σ	nested-logit within-nest correlation parameter
α	price coefficient
$s_{i,t}$, $s_{i\|g,t}$	unconditional / within-nest conditional share
M_t, $q_{i,t} = M_t s_i(p_t, \omega_t)$	market size; quantity (share $\times$ market size) — M_t cancels from the pricing FOC
$c(\omega_{i,t})$	marginal cost as a function of own state
FC	fixed per-period operating cost (distinct from the fringe, F)
$\pi_i^*(\omega_t)$	equilibrium static profit from the pricing subgame
ϕ_i	exit scrap value (matches Ericson–Pakes' own ϕ)
x_e	sunk entry cost (matches Ericson–Pakes' own x_e)
$\lambda^{\text{exit}}, \lambda^{\text{entry}}$	0/1 exit / entry indicators implied by the value-function cutoffs (13)/(15) — not η (EP95's own η' is an unrelated aggregate shock) or χ (claimed by the exit-decision indicator in the Doraszelski–Pakes–Satterthwaite restatement), and not σ (the nesting parameter)
$G_e(\cdot)$	entrant's initial-state distribution, $\omega_{e,0} \sim G_e(\cdot)$
β	discount factor
$V_i(\cdot)$	incumbent value function